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Impact of thermal-acoustic exposure on human performance of healthy adults: simulation of traffic noise and thermal conditions in the urban environment

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Occupants in many urban and occupational environmental settings are simultaneously exposed to environmental traffic noise and heat stress. These combined exposures may have adverse effects on physiological functions and cognitive performance. This study aimed to assess the effects of concurrent exposure to traffic noise and thermal stress under controlled laboratory conditions. In this crossover experimental study, 80 healthy participants (mean age: 25.3 ± 2.8 years) were exposed to four exposure conditions: (1) control (comfort), (2) thermal (WBGT = 30.62 °C), (3) traffic noise (75 dB_A), and (4) combined exposure (WBGT = 30.62 °C and75 dB_A). Physiological parameters including heart rate, blood pressure, tympanic temperature, and mean skin temperature were recorded. Cognitive performance was assessed using the Stroop test (for selective attention) and n-back test (for working memory). Repeated-measures ANOVA was used for statistical analysis. Exposure to combined heat and traffic noise produced the most pronounced physiological and cognitive alterations among all test conditions. Core temperature rose significantly during noise ($p = 0.008$) and combined exposure ($p = 0.007$). Diastolic blood pressure increased under thermal ($p = 0.001$) and combined conditions ($p < 0.001$). Heart rate also elevated during combined exposure ($p = 0.032$). CPT reaction time lengthened significantly across conditions ($p = 0.001$), with the thermal and combined situation more than control (post hoc $p < 0.05$, $p < 0.01$). All n-back outcomes showed statistically significant differences across exposure conditions at levels 2 and 3, and for incorrect responses and mean reaction time at level 1 ($p < 0.001$). Simultaneous exposure to traffic noise and thermal stress amplifies both physiological strain and cognitive impairment compared to single exposures. These findings highlight the importance of integrated exposure assessment and risk mitigation strategies in occupational environments, especially in warm and noisy settings.

Keywords Traffic noise, Themal stress, Environmental exposure, WBGT, Performance

Exposure to noise pollution, particularly traffic noise, impairs cognitive functions, leading to reduced accuracy in brain activities, compromised cognitive coordination, and disruptions in communication and memory. Traffic noise is recognized as an environmental stressor that degrades cognitive performance during mental processing and elevates the risk of errors. For instance, complex cognitive tasks requiring high vigilance and rapid reaction times—such as those critical for emergency response during driving—are significantly impaired under traffic noise exposure[1,2]. This attentional impairment substantially diminishes individual performance and, with prolonged exposure, may elevate the incidence of traffic accidents[3]. Exposure to environmental noise (e.g., traffic) induces detrimental changes in precision, attention, and processing speed, concurrently heightening psychological stress[4]. As an environmental stressor, traffic noise contributes to public health issues among urban populations, including physiological and psychological disorders. Physiologically, it exerts deleterious effects on cardiovascular activity, elevates blood pressure, increases myocardial infarction risk, and disrupts sympathetic

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nervous system function. Additionally, it induces endocrine system dysfunction, elevates serum lipid levels, and increases norepinephrine secretion[5]. Transportation-related noise pollution is further established as a significant modifier of cognitive and mental performance in exposed populations[6]. Supporting this, Choi et al. (2015) demonstrated elevated stress levels in subjects exposed to traffic noise through electroencephalographic (EEG) analysis. Research indicates that increased alpha wave amplitude (μV) correlates with enhanced cognitive performance, whereas attenuation reflects heightened psychological stress[7].

Thermal stress represents another significant environmental stressor impairing cognitive performance in both indoor and outdoor occupational settings. Heat exposure may originate from industrial processes or ambient climatic conditions. Beyond inducing physiological disorders and heat-related illnesses, thermal stress triggers adverse cognitive responses that degrade human performance[8]. As a physical hazard in occupational studies, heat stress not only causes physiological impairments but also compromises cognitive functions—specifically disrupting decision-making processes and the cognitive transition from simple to complex tasks[8]. Elevated core body temperature appears to initiate a physiological cascade that alters mental states and impairs decision-making capacity[9]. Empirical evidence indicates generalized reductions in processing speed and accuracy under elevated ambient temperatures. Critically, thermal stress may potentiate noise-induced health effects through synergistic interactions[10]. Notably, methodological limitations in single-stressor cognitive models have drawn increasing criticism in recent years. Studies examining isolated stressors exhibit significant contradictions, methodological shortcomings, and interpretive limitations[8,11]. Consequently, researchers now advocate investigating combined environmental exposures as a more valid approach to assess physiological and cognitive health impacts[12]. Guan et al. (2020) demonstrated that speech intelligibility across varying sound pressure levels (SPL) is significantly modulated by ambient thermal conditions[13]. In a simultaneous noise-climate exposure study, Abbasi et al. (2020) observed that high-intensity noise exposure (compared to low-intensity) reduced alpha wave amplitude (μV) across all temperature regimes[14]. This amplitude attenuation confirms cognitive performance degradation during concurrent exposure task execution. Chen et al. conducted controlled experiments in a climate chamber, exposing 14 healthy young adults to combined noise and thermal stimuli. They concluded that heat stress exacerbates noise-induced temporary threshold shift (TTS)[15]. Furthermore, studies revealed that thermal comfort was significantly compromised by concurrent noise exposure during combined stressor conditions[16,17].

Over recent decades, the impact of environmental factors on human comfort, health, and performance has been extensively examined in both laboratory and field settings. Although a growing number of studies have adopted multi-domain approaches and investigated combined environmental stressors, including interactions between thermal conditions and noise[18,19], evidence regarding their joint effects on human responses remains heterogeneous. While some investigations have reported significant interactions between noise and thermal stress on physiological, perceptual, or cognitive outcomes particularly under moderate-to-high exposure levels[13,14,20], Also others have not observed clear interaction effects, potentially due to differences in sound pressure levels, temperature ranges, task demands, or controlled office-like settings[21,22]. This inconsistency indicates that the physiological and cognitive consequences of concurrent thermal–acoustic exposure are not yet fully understood and merit further systematic study. From a methodological perspective, prior research has often focused on limited sets of outcome measures, frequently emphasizing subjective ratings of comfort, annoyance, or performance (e.g., thermal sensation votes, noise annoyance scales), and only partially integrating objective physiological and cognitive indicators during simultaneous exposure conditions[13]. This constrains the ability to elucidate mechanistic pathways and link perceived states with measurable functional impairment. Unlike prior work[23], which primarily examined thermal comfort interactions in general indoor environments using subjective assessments among healthy adults, the present study investigates traffic-derived noise combined with heat stress on a broader array of objective physiological and task-based cognitive performance measures within a controlled simulation replicating real urban traffic and climatic conditions of Sari, Iran.

Materials and methods

Study design and participants

In this within-subjects crossover experimental study with repeated measures design, 80 healthy university students (40 males, 40 females; mean age 26.9 ± 2.8 years, range 22–28 years) from Mazandaran University of Medical Sciences were exposed to all four experimental conditions (control comfort, thermal WBGT = 30.6 °C, traffic noise 75 dBA, combined exposure) on separate days (maximum one scenario per day) with randomized scenario sequence to eliminate order effects. All participants were from the same academic cohort, ensuring homogeneity in age, education, and lifestyle factors. Inclusion criteria required: (1) self-reported absence of systemic disease or clinical disorders, (2) normal audiometric thresholds (≤ 25 dB HL, 0.5–4 kHz), and (3) visual acuity ≥ 20/25 (Snellen E-chart). Pre-test protocols mandated ≥ 8 h of sleep and abstinence from caffeinated beverages (tea/coffee/chocolate) ≥ 4 h pre-assessment. Exclusion criteria encompassed: (1) psychiatric conditions or psychoactive medication use, (2) tobacco/alcohol consumption within 24 h, (3) study withdrawal, or (4) incomplete data. Voluntary participation with written informed consent was obtained following institutional ethics board approval. In this study, incentives were provided to enhance participant motivation and ensure diligent and accurate execution of the experiments, despite the time-intensive and sometimes repetitive nature of the testing procedures. Consequently, all participants successfully completed the experimental sessions. Prior to each experiment, informed consent was obtained following a comprehensive explanation of the study protocol. Moreover, participants were allowed to choose their preferred day and time for test administration. Participants wore standardized light summer clothing ensembles (Light office clothing Clo ≈ 0.5–0.7) to minimize inter-individual thermal resistance variation, consistent with moderate office work attire (metabolic rate ~ 1.2 met)[24].

Procedure

This study implemented four experimentally controlled exposure scenarios. (1) control (comfort), (2) thermal exposure (WBGT = 30.62 °C), (3) traffic noise exposure (75 dB_A), and (4) combined or simultaneous exposure (WBGT = 30.62 °C C and75 dB_A). Background noise levels were measured in the acoustically treated listening room during quiet control (no traffic noise stimulus) and thermal-only exposure conditions using the Casella CEL-450 sound level meter (Class 1, IEC 61672-1:2013) at ear height (120 cm). Mean LAeq values were 35.5 ± 2.2 dBA (quiet control) and 27 ± 1.1 dBA (thermal-only, WBGT = 30.62 °C), confirming effective acoustic isolation (< 30 dBA) well below the 75 dBA traffic noise stimulus across all experimental scenarios. Each scenario was administered to participants on separate days (maximum one scenario/day), with scenario sequence randomized across subjects to eliminate order effects (i.e., exposures did not follow sequential Scenario 1→4 progression). Figure 1 presents the simulated exposure protocol schematic:

This study implemented four experimentally controlled exposure scenarios in a single acoustically treated environmental chamber (210 × 310 × 450 cm, L×W×H) with integrated thermal and acoustic control systems: (1) control (comfort: thermoneutral and background noise), (2) thermal exposure only (WBGT = 30.62 °C, background noise < 40 dBA), (3) traffic noise exposure only (75 dBA LAeq, thermoneutral WBGT), and (4) combined exposure (WBGT = 30.62 °C + 75 dBA LAeq simultaneously).

For each scenario, the chamber was specifically configured by activating the relevant thermal and acoustic subsystems while maintaining all other parameters constant. Each scenario was administered to participants on separate days (maximum one scenario/day), with scenario sequence randomized across subjects to eliminate order effects. The experimental chamber featured controlled climatic conditions including dry-bulb temperature, wet-bulb temperature, relative humidity, and 300 lx illumination. The 210 × 310 × 450 cm (L×W×H) soundproofed enclosure contained an administrative workstation with an ergonomic office chair, work desk, and visual display terminal for functional performance testing. Interior surfaces (walls and floor) were constructed of wood, with the chamber providing full acoustic isolation (Fig. 2). Also chamber conditions were continuously monitored for air quality (CO2 < 800 ppm via indoor air quality meter, PM2.5 < 15 μg/m³), relative humidity (40–60%), and air velocity (0.2–0.3 m/s via anemometer), ensuring confounds did not influence physiological or cognitive outcomes beyond target WBGT and noise exposures.

Reproduction and anechoic chamber (traffic noise survey, produce and playback)

Prior to generating and presenting urban traffic noise (specific to Sari) to participants, field measurements of traffic noise across Sari's streets were required. Measurements were conducted on Saturdays (first workday) from 8:00–11:00 AM to replicate typical traffic exposure conditions and align with peak vehicle flow in congested urban corridors (12 measurement points). Based on community noise reports, eleven high-traffic locations were selected for 3-h recording sessions. Traffic noise was captured during summer under controlled conditions: sunny weather, windless, dry asphalt, with vehicles maintaining approximately 30 km/h. Recording utilized a SONY PCM-D50 portable recorder with Sennheiser ME 67 shotgun microphone and dynamic stereo headphones for monitoring. Concurrently, a Casella CEL-450 sound level meter (Class 1 accuracy per IEC 61672-1:2013) with A-weighting network measured equivalent continuous sound levels (LAeq) and frequency spectra. The device was positioned at 120 cm height and 2 m from road edges during 30-minute sampling intervals. Measurement sites avoided reflective surfaces (vegetation, buildings) to minimize acoustic interference. Laboratory post-

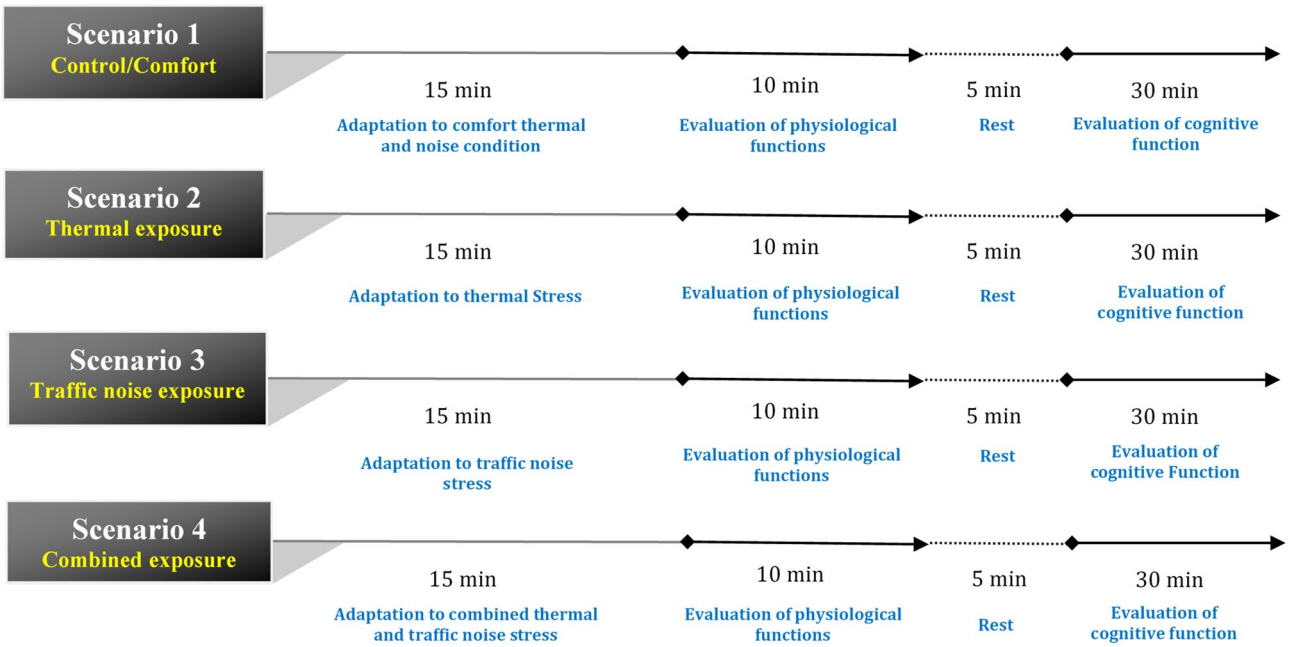


Fig. 1. Simulation model of exposure scenarios.

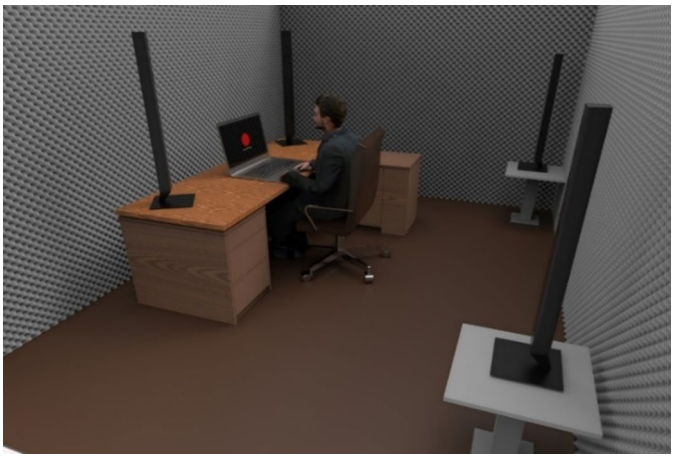


Fig. 2. The schematic experimental chamber to simulate exposures.

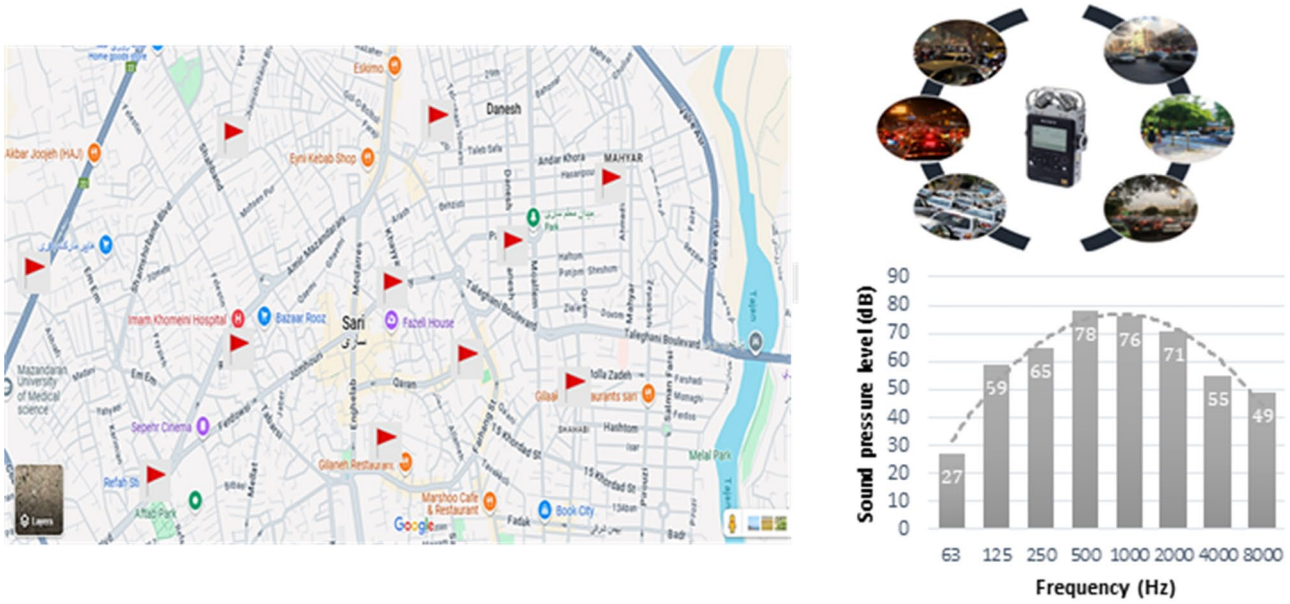


Fig. 3. Geographic distribution of soundscape recording sites and corresponding results of A-weighted octave-band frequency analysis of the captured traffic noise, which was subsequently reproduced in an acoustically treated listening room.

processing involved editing raw recordings using Cool Edit Pro (Adobe Systems, San Jose, CA, USA) to create standardized traffic noise clips matching field-measured exposure levels (75 dBA LAeq). Mono field recordings captured with the Sennheiser ME 67 shotgun microphone were upmixed to a four-channel configuration for playback through the four loudspeakers shown in Fig. 2 (two front, two rear speakers positioned at ear height, 1.5 m listener distance). This standard stereo duplication with equal panning across channels preserved the original spectral content and temporal dynamics of Sari traffic noise while providing balanced spatial distribution and adequate immersion within the acoustically treated listening room. No advanced spatialization (e.g., ambisonics or HRTF) was applied, as the study focused on overall noise level effects rather than directional sound localization[25]. Final playback to participants occurred through two loudspeakers in an acoustically treated, sound-attenuated listening room, where background noise levels were maintained at LAeq < 40 dBA (35.5 ± 2.2 dBA in control conditions), ensuring clear audibility and signal-to-noise ratio > 45 dB for the standardized 75 dBA traffic noise stimulus (Fig. 3).

Reproduction of thermal stress (thermal stress survey, produce)

To simulate heat stress from Sari's summer climatic conditions, the environmental chamber regulated dry-bulb temperature (Ta), wet-bulb temperature (Tw), and black globe temperature (Tg) to achieve target WBGT =

30.62 °C, calculated from historical meteorological data (Iran Meteorological Organization). Tg (the radiant temperature component reflecting mean radiant heat load) was empirically estimated using validated models[26,27] from environmental variables (air temperature, humidity, wind speed) and independently simulated via electric heaters with precision controllers distributed across chamber surfaces to achieve spatial homogeneity. Required thermal variables were generated by electric heaters and humidity control systems calibrated to Sari thermal scenarios, with air velocity maintained at 0.2–0.3 m/s. Chamber walls, ceiling, and floor featured thermal/moisture insulation with sealed joints. A REED R6200 WBGT monitor (Taiwan) continuously verified all three parameters (Ta, Tw, Tg) at torso height (120 cm) during experimental phases, ensuring WBGT stability within ± 0.5 °C of Sari summer conditions. While short-wave solar radiation was not directly simulated (consistent with standard indoor WBGT protocols), the controlled Tg effectively replicated the integrated radiant thermal load of outdoor urban environments.

Human performance

Physiological parameters

Physiological parameters: As illustrated in Fig. 1 (exposure protocol schematic), prior to activation of each experimental scenario, participants underwent a 15 minutes' rest period in the environmental chamber (thermoneutral and background noise: comfort conditions) for initial acclimatization, followed by 5-minute baseline monitoring of heart rate, core body temperature, and blood pressure. Stressors (thermal and/or traffic noise) were then activated, and these parameters were reassessed during minutes 1–5 (initial exposure phase) and the final 5 min of each 30-minute exposure session, consistent with the timeline shown in Fig. 1. Core temperature was monitored tympanically using a Beurer FT78 infrared thermometer. Blood pressure (systolic/diastolic) and heart rate were recorded oscillometrically using a Beurer BM16 digital sphygmomanometer. This standardized protocol—baseline under thermoneutral/quiet conditions followed by scenario-specific stressor activation—was implemented consistently across all four experimental scenarios.

Cognitive performance

Sustained attention and vigilance were evaluated using the Continuous Performance Test (CPT), while working memory was assessed through a three-level n-back task, serving as cognitive performance measures under different exposure conditions[28,29]. The CPT requires continuous monitoring of rapidly presented visual stimuli and fast discrimination between target and non-target items, yielding omission errors, commission errors, and reaction time as indices of attentional lapses, response inhibition failures, and processing speed, respectively. These processes are directly relevant to real-world street and traffic behaviour, where individuals must continuously scan the environment, detect critical events (e.g., pedestrians, vehicles, signals), and respond quickly and accurately to avoid errors. The n-back task engages online maintenance and updating of information under increasing cognitive load (levels 1–3), providing reaction time and accuracy measures that reflect working memory capacity and executive control. Such functions support complex decision-making in dynamic environments, including route selection, adaptation to changing traffic conditions, and integration of multiple concurrent stimuli. Together, CPT and n-back therefore capture key cognitive domains that underpin safe performance in urban traffic contexts and have been widely used in prior human studies investigating environmental effects on attention, vigilance, and working memory. In the CPT, 150 numbers appear at specified time intervals, with one stimulus designated as the target stimulus. Participants were instructed to respond by pressing the corresponding key on the keyboard as quickly as possible upon presentation of the target numbers. The measured variables in this study include omission errors, commission errors, and reaction or response time (milliseconds). The n-back (as a working memory task) is a cognitive performance assessment related to executive functions. This test has three complexity levels, and all three levels—first (n1), second (n2), and third (n3)—were utilized in this research. Level 1 procedure: Numbers 1 to 9 appear randomly on the screen for a short duration; participants must remember the first number and wait for the second number. After the second number appears, if it matches the previous number, they press the (correct) key as quickly as possible; if different, they press the (incorrect) key, continuing this pattern throughout the test. Level 2 procedure follows the same method as Level 1 but compares the first number with the third, second with fourth, and so forth. In this test, mean reaction time and correct responses can be extracted. Level 3 procedure follows the same method as Level 1 but compares the first number with the fourth, second with fifth, and so forth. In this test, mean reaction time and correct responses can be extracted[30–32]. Cognitive performance assessments were conducted for each participant under four scenarios: (1) thermal comfort conditions, (2) heat stress exposure, (3) traffic noise exposure, and (4) simultaneous heat stress and traffic noise exposure. Notably, to standardize procedures and prevent laboratory errors due to peak cognitive performance hours (approximately 8:00–12:00), all testing occurred within this fixed daily timeframe (8:00–12:00).

Statistical analysis

The data were initially screened for distribution characteristics and potential violations of assumptions. We assessed the normality of the dependent variables across the four exposure conditions using the Shapiro–Wilk test, along with visual inspections of histograms and Q-Q plots. All primary physiological and cognitive outcomes exhibited approximately normal distributions within each condition. Since all 80 participants completed each of the four exposure scenarios, the dataset forms a fully balanced within-subjects repeated-measures design, making it suitable for repeated-measures ANOVA. We examined the sphericity assumption for the within-subject factor (exposure condition), and where necessary, we applied standard corrections (e.g., Greenhouse–Geisser adjustment) to the degrees of freedom for the F-tests. Repeated-measures ANOVA was employed as the primary inferential method for normally distributed outcomes, while the non-parametric Friedman test was used when normality was not satisfied. When the global tests yielded significant results, we conducted post hoc

Variable	Minimum	Maximum	Mean
Noise (dB)	74.24	76.66	75.45
Dry temperature (°C)	29.70	34.90	32.30
Wet temperature (°C)	25.00	27.00	26.00
Globe temperature (°C)	29.50	35.50	32.50
WBGT (°C)	30.00	31.25	30.62

Table 1. Measured Exposure Levels of environment factors.

Variable	Without exposure (comfort)		Beginning of exposure		End of exposure		*P* value
	Mean	SD	Mean	SD	Mean	SD	
Traffic noise exposure							
Systolic BP (mmHg)	112.5	10.8	111.3	10.2	110.8	12.4	0.452
Diastolic BP (mmHg)	69.8	9.1	70.8	8.8	70.8	8.9	0.761
Heart rate (bpm)	83.1	12.2	84.9	11.9	83.4	11.9	0.249
Core temperature (°C)	35.3	0.7	35.4	0.6	35.4	0.7	0.008**
Thermal exposure							
Systolic BP (mmHg)	111.9	9.8	111.3	8.4	112.1	9.9	0.705
Diastolic BP (mmHg)	70.3	7.2	73.6	7.9	73.7	7.8	0.001**
Heart rate (bpm)	80.9	10.7	81.8	8.9	81.9	10.1	0.717
Core temperature (°C)	34.6	0.4	34.7	0.4	34.8	0.4	0.057
Combined exposure							
Systolic BP (mmHg)	111.3	10.6	114.8	10.6	114.8	10.7	0.053
Diastolic BP (mmHg)	69.8	6.8	75.4	7.4	75.1	7.9	0.00**
Heart rate (bpm)	80.9	9.9	83.3	9.7	82.9	9.4	0.032*
Core temperature (°C)	34.7	0.4	34.8	0.5	34.8	0.4	0.007**

Table 2. Physiological parameters across exposure conditions. *P-values from RM-ANOVA main effect (beginning vs. end within each condition). *$p < 0.05$, **$p < 0.01$.

pairwise comparisons between the exposure conditions using the Wilcoxon signed-rank test, with a significance level set at 0.05. All statistical analyses were performed using SPSS (Version 25.0, IBM Corp., Armonk, NY, USA).

Results

Participants were exposed to traffic noise at 75.45 ± 1.21 dBA (range: 74.24–76.66 dBA). Other environmental exposure levels are presented in Table 1.

Physiological parameters (Table 2) showed significant within-condition changes (main effects): core temperature increased during traffic noise exposure ($p = 0.008$), diastolic blood pressure (DBP) during thermal exposure ($p = 0.001$), and DBP ($p < 0.001$), heart rate ($p = 0.032$), and core temperature ($p = 0.007$) during combined exposure. Pairwise comparisons between exposure conditions showed that diastolic blood pressure (DBP) was significantly higher in the combined heat–noise condition than in traffic noise alone at both the beginning ($p = 0.010$) and end ($p = 0.013$) of exposure, and higher in combined than thermal exposure at the beginning ($p = 0.033$). Systolic blood pressure exhibited a smaller but consistent trend, with a significant difference for combined versus traffic noise at the end of exposure ($p = 0.018$) and a near-significant difference at the beginning ($p = 0.053$). These results indicate that the combined condition produces greater circulatory strain than either single stressor, particularly for DBP, even though the magnitude of between-condition differences is modest. The absolute differences in DBP between combined and traffic-only conditions were approximately 4–6 mmHg at both time points, suggesting a small but physiologically relevant increment in vascular load under dual exposure.

Table 3 presents the results of pairwise comparisons between the different exposure conditions for systolic and diastolic blood pressure, heart rate, and core body temperature (p-values for between-condition tests). As shown, the p-values indicate that the combined heat-and-noise condition produced significantly higher values for some cardiovascular indices, particularly diastolic blood pressure, compared with noise-only or heat-only exposure, whereas other differences remained small and non-significant. This pattern suggests that simultaneous exposure to heat and traffic noise can increase cardiovascular load more than each stressor alone, although the magnitude of these effects is generally in the mild to moderate range.

In Table 4, the reaction time showed significant increases across exposure conditions ($p = 0.001$ overall ANOVA), with post-hoc analyses revealing thermal exposure (455.00 ± 43.17 ms) > no-exposure (432.71 ± 46.06

Variables	Comparison of exposures		
	Thermal vs. traffic noise	Combined vs. traffic noise	Combined vs. thermal
Systolic BP (mmHg)			
Beginning of exposure	0.134	0.018	0.851
End of exposure	0.063	0.013	0.528
Diastolic BP (mmHg)			
Beginning of exposure	0.033	0.010	0.102
End of exposure	0.308	0.013	0.174
Heart rate (bpm)			
Beginning of exposure	0.362	0.669	0.461
End of exposure	0.584	0.214	0.579
Core temperature (°C)			
Beginning of exposure	0.907	0.674	0.922
End of exposure	0.170	0.921	0.390

Table 3. Pairwise comparisons of systolic and diastolic blood pressure, heart rate, and core temperature between exposure conditions (values are p values for between-condition tests). Main effects reflect within-condition changes; between-condition interactions. Values are p-values for pairwise comparisons between exposure conditions at the beginning and end of exposure. $^{*}p < 0.05$, $^{**}p < 0.01$ (statistically significant).

Variable	Condition	Mean ± SD	*P* value
Commission errors	Without exposure	2.29 ± 0.66	0.544
	Traffic noise exposure	2.51 ± 0.87	
	Thermal exposure	2.60 ± 0.91	
	Combined exposure	2.60 ± 0.77	
Omission errors	Without exposure	2.39 ± 0.28	0.280
	Traffic noise exposure	2.39 ± 0.28	
	Thermal exposure	2.60 ± 0.56	
	Combined exposure	2.63 ± 0.62	
Correct responses	Without exposure	149.49 ± 0.82	0.183
	Traffic noise exposure	149.26 ± 0.98	
	Thermal exposure	149.09 ± 1.07	
	Combined exposure	149.09 ± 1.01	
Reaction time (ms)	Without exposure	432.71 ± 46.06[a]	0.001
	Traffic noise exposure	445.97 ± 27.23[ab]	
	Thermal exposure	455.00 ± 43.17[b]	
	Combined exposure	461.57 ± 51.53[b]	

Table 4. Cognitive attention parameters (CPT) across exposure conditions. *Superscript letters denote significant post hoc pairwise differences ($p < 0.05$). Groups sharing at least one common letter (e.g., [a] or [b]) do not differ significantly from each other; groups with no letters in common differ significantly.

ms; $p < 0.05$) and combined exposure (461.57 ± 51.53 ms) > no-exposure ($p < 0.01$). Thermal exposure demonstrated broader trends toward impairment (increased commission/omission errors, reduced correct responses), while combined exposure primarily prolonged reaction time, suggesting thermal stress as the dominant driver of attentional slowing with additive noise effects on processing speed. Error rates showed non-significant trends consistent with this pattern.

All performance metrics at levels 2 and 3 (incorrect responses, correct responses, % accuracy, mean reaction time) and level 1 incorrect responses and mean RT showed significant differences across exposure conditions (all $p < 0.001$). Post-hoc analyses revealed consistently impaired performance under thermal and combined exposures compared to no-exposure control, with combined conditions showing numerically worse outcomes than thermal alone across most indices (e.g., level 3% accuracy: combined 77.86 ± 19.01% vs. thermal 82.94 ± 13.53% vs. control 70.54 ± 16.47%). The mean reaction times were shortest in combined exposure at Level 3 n-back (e.g., Level 3: 590.57 ± 182.74 ms vs. control 721.83 ± 213.34 ms). All n-back outcomes showed statistically significant differences across exposure conditions at levels 2 and 3, and for incorrect responses and mean reaction time at level 1 (all $p < 0.001$). At level 1, incorrect responses were slightly higher under thermal and combined exposures compared with traffic noise and no-exposure, while mean reaction time tended to be longest under combined exposure, indicating an early increase in cognitive load under heat and combined stress. At levels 2 and 3, thermal and combined exposures generally produced more pronounced impairments in

Variable	Without exposure (Mean ± SD)	Traffic noise exposure (Mean ± SD	Thermal exposure (Mean ± SD	Combined exposure (Mean ± SD	*P* value
Level 1 n-back					
Incorrect responses	2.67 ± 2.90	2.01 ± 2.73	2.86 ± 3.47	2.46 ± 4.23	0.027
Correct responses	114.94 ± 3.68	116.83 ± 2.80	112.14 ± 18.04	114.09 ± 7.88	0.262
% Correct responses	95.57 ± 3.06	97.23 ± 2.28	93.40 ± 15.06	95.03 ± 6.52	0.278
Mean reaction time	588.43 ± 138.11[a]	488.63 ± 167.94[b]	478.11 ± 122.63[b]	521.04 ± 125.86[ab]	< 0.001
Level 2 n-back					
Incorrect responses	3.30 ± 7.46[a]	2.40 ± 8.35[b]	2.09 ± 7.10[b]	2.21 ± 8.64[b]	< 0.001
Correct responses	88.03 ± 16.67[a]	99.46 ± 11.99[b]	101.80 ± 12.52[b]	98.03 ± 14.74[b]	< 0.001
% Correct responses	73.31 ± 13.87[a]	82.77 ± 9.96[b]	84.69 ± 10.46[b]	81.60 ± 12.32[b]	< 0.001
Mean reaction time	732.37 ± 196.91[a]	586.91 ± 176.01[b]	539.06 ± 148.43[b]	593.80 ± 188.58[b]	< 0.001
Level 3 n-back					
Incorrect responses	3.37 ± 12.89[a]	2.14 ± 11.17[b]	1.99 ± 9.98[b]	2.50 ± 12.29[b]	< 0.001
Correct responses	84.83 ± 19.85[a]	95.94 ± 16.46[b]	99.60 ± 15.83[b]	93.54 ± 22.83[b]	< 0.001
% Correct responses	70.54 ± 16.47[a]	79.83 ± 13.75[b]	82.94 ± 13.53[b]	77.86 ± 19.01[b]	< 0.001
Mean reaction time	721.83 ± 213.34[a]	614.09 ± 197.36[b]	547.86 ± 159.69[b]	590.57 ± 182.74[b]	< 0.001

Table 5. Working memory performance (n-back test) across exposure conditions. Superscript letters indicate significant post hoc pairwise differences ($p < 0.05$). Within each row, conditions sharing the same letter do not differ significantly; conditions with different letters differ significantly.

working memory than the noise-only condition. Together, these results suggest that heat and combined thermal–noise exposure impose a greater working memory burden than noise alone, but that the relative impact of each exposure condition can differ somewhat across task difficulty levels and specific performance indices (Table 5).

Discussion

Core temperature demonstrated statistically significant changes during both traffic noise exposure and combined traffic noise-heat stress exposure across experimental conditions. Hemmatjou et al. similarly reported elevated core temperature under concurrent heat and noise exposure[33,34]. High noise levels induce physiological stress, triggering adrenaline secretion and peripheral vasoconstriction that elevates blood pressure. Additionally, noise exposure alters heart rate, reduces cardiac output, and increases respiratory rate, ultimately leading to body temperature fluctuations[35].

Statistically significant correlations were identified between initial and final systolic blood pressure measurements during combined traffic noise-heat stress exposure and corresponding initial and final systolic blood pressure measurements during traffic noise exposure alone. In a 2015 study by Dehghan et al. examining the effects of simultaneous noise and heat exposure on blood pressure changes in men, combined exposure resulted in a slight increase in blood pressure - consistent with the current study's findings[36]. Sepehri et al. (2019) similarly reported significant differences in mean systolic blood pressure changes across heat levels during exposure to annoying noise (55–75 dB), aligning with our findings of BP elevation under thermal-acoustic stress. Like our laboratory study, theirs used a climatic chamber with healthy male students to examine combined effects, though their focus was low air temperatures (14–22 °C) rather than our WBGT = 30.6 °C summer simulation. Both demonstrate cardiovascular sensitivity to physical stressor interactions, supporting the physiological plausibility of our urban exposure model[37].

Significant associations were observed for diastolic blood pressure during both heat stress exposure and combined exposure to traffic noise and heat stress. Increased heat stress elevates core body temperature, leading to heat-related disorders such as heat cramps, heat exhaustion, and heat stroke, consequently altering blood pressure levels[38]. Concurrently, noise exposure triggers short-term physiological responses via the autonomic nervous system resulting in elevated blood pressure, heart rate, and peripheral vasoconstriction. In a similar study by Dehghan et al. (2015) examining the combined effects of noise and heat on blood pressure in males, concurrent exposure caused a slight increase in blood pressure, aligning with the present findings[36].

Diastolic blood pressure during combined traffic noise and heat stress exposure significantly differed from exposure to traffic noise alone. Post-exposure diastolic pressure under combined conditions showed a significant increase compared to isolated noise exposure. This phenomenon occurs because heat exposure initially alters intravascular fluid volume, subsequently affecting blood pressure. Subsequent sympathetic stress responses) and endocrine activation elevate blood pressure, heart rate, stress hormones, and notably oxidative stress, which contributes to vascular dysfunction and cardiovascular risk factors (e.g., hypertension, blood viscosity, hyperglycemia, coagulation activation)[39]. Previous studies corroborate these results, demonstrating significant blood pressure elevation following noise exposure[40].

Heart rate variability exhibited significant associations during combined noise and heat stress exposure. This may stem from synergistic effects of traffic noise and heat on stress pathways and endocrine function, elevating both blood pressure and heart rate[39]. A closely related study by Wang et al. on Chinese steelworkers exposed to noise, heat, or both revealed that physical stressors affect heart rate, with combined exposure causing the most pronounced increase[41]. Conversely, Jafari et al. reported that heart rate changes during isolated or combined noise/heat exposure were less pronounced than systolic/diastolic blood pressure alterations[42].

Working memory performance (as measured by total incorrect responses across Levels 1–3) showed significant associations with different exposure conditions. Traffic noise exposure increased error rates, indicating impaired attentional capacity. This may stem from noise-induced stress, which reduces alpha wave amplitude (μV) during exposure, thereby increasing mental workload and diminishing cognitive performance[7]. *Monteiro et al.* demonstrated that noise exposure significantly elevates error rates and reduces attention levels[43], while *Tzivian et al. (2017)* reported similar findings for sustained attention[44]. Contrary to Level 1 results, error rates in Levels 2 and 3 of the same tests decreased significantly during exposure to traffic noise, heat stress, or combined conditions compared to the no-exposure baseline. This aligns with *Vasmatzidis et al. (2002)*, who found no significant link between noise/heat exposure and cognitive performance in exposed vs. control groups[45]. In this regard, the pre-test training effect, which repeated task practice minimised gross estimation errors, exists, and another factor is the Educational attainment influence, that Participants' education levels (high school diploma to doctoral degrees) correlated with enhanced cognitive resilience. Most studies reporting contradictory findings involved populations with < 12 years of formal education.

Among other findings of this study, a significant association can be mentioned for the cognitive factor of working memory in the variable of correct response percentage in levels 2 and 3 with different exposure conditions; that is to say, the percentage of correct responses during exposure to traffic noise, heat stress, or combined exposure to traffic noise and heat stress compared to the no-exposure condition has increased. Therefore, contrary to the expected result, exposure to various intervention factors has no effect on the percentage of correct responses in different levels of the mentioned test. McDonnell et al. (2021) in a study investigating the relationship between road traffic noise and cognitive performance also found no evidence indicating an association between noise exposure and memory with processing speed, and this may be due to small sample size[46]. In this research, similar to the study by Golbabaei et al. (2014), it was not possible to stratify individuals based on education level, which could be one of the reasons for the difference in the results of this study compared to some other researches[47]. Also, this matter could be due to the pre-test training and repetition of previous test stages, which leads to individuals having more mastery and higher skill, and as a result, the percentage of correct responses in each stage increases compared to the no-exposure stage.

The obtained findings showed that the cognitive factor of working memory in the variable of mean response time of all levels 1–3 had a significant association with different exposure conditions; that is to say, the mean response time during combined exposure to traffic noise and heat stress compared to the heat stress condition has increased. Sepehri and his colleagues also encountered such similar results in their research in 2017. They found that the combined effect of noise and air temperature on reaction time was greater than the effect of each of them separately[37]. Abbasi and colleagues also in 2018 [1397] found that the combined effects of noise and air temperature are greater than the effect of each of them alone[14].

Regarding the present study's data concerning mean response time in levels 1–3, there is a significant decrease; that is to say, the mean response time during exposure to traffic noise, heat stress, or combined exposure to traffic noise and heat stress compared to the no-exposure condition has decreased. Smith reached a similar result in this field that the combined effects of noise and heat on cognitive functions are not related to each other. Also, in studies by Hygge and Yang, no interaction between noise and heat stress was reported[21,22]. Also, this matter could be due to the pre-test training and repetition of previous test stages, which leads to individuals having more mastery and higher skill, and as a result, the mean response time in each stage decreases compared to the next stage.

Regarding the present study's data, reaction time in the heat stress condition has shown a significant increase because heat causes reduction of blood supply to the brain and leads to reduction of mental performance in individuals[48,49]. These findings match the results of Gaoua's study in 2011. He found that with increasing heat, mental performance of individuals decreases[50]. Also, Habibi and colleagues found that heat through reduction of precision during cognitive activities causes reduction of mental performance in individuals[51].

The results of the present study indicate that traffic noise and heat stress have a reciprocal interaction and can increase each other's adverse effects. Because due to heat, blood supply to the brain is reduced and causes reduction of mental performance in individuals[48]. Also, due to traffic noise, because of creating high stress during exposure, mental workload increases and mental performance of the brain decreases[7]. In a study that Chen and colleagues conducted, also such similar results were obtained and it was determined that combined exposure to noise and heat stress causes impairment in memory[15]. Also, in the study by Abbasi and colleagues, combined effects of noise and air temperature on working memory were greater than the effect of each of them alone[14]. In the case that Zamanian and Smith in this regard found that the combined effects of noise and heat on cognitive functions are not related to each other[52,53].

This study was conducted in a controlled acoustic chamber that allowed precise simulation of traffic noise (based on field measurements) with uniform distribution and specific intensity variation, enabling isolation of thermal-acoustic effects on continuous cognitive performance. The ability to independently control heat stress parameters alongside realistic urban noise playback represents a key methodological strength. However, several limitations should be noted. The sample consisted of young, healthy university students with likely prior awareness of heat and noise effects due to their medical sciences background, which may have influenced subjective responses. This homogeneous group does not represent vulnerable populations, such as older outdoor workers, and thus limits generalizability to occupational settings. The laboratory environment, while ecologically valid for controlled urban simulation, cannot fully capture real-world factors like prolonged exposure, physical workload, or individual acclimatization. Additional other affectiong variables such as personality traits, acoustic characteristics, and thermal radiation properties may also influence outcomes. The study was not powered for subgroup analyses (e.g., gender interactions), and future research should validate these findings in diverse, high-risk cohorts.

Conclusion

This controlled experimental study provides robust evidence that simultaneous exposure to traffic noise and thermal stress significantly compromises both physiological homeostasis and cognitive functioning in young adults. While individual exposures to either traffic noise or heat stress elicited only modest physiological alterations, the combined exposure produced synergistic effects, leading to marked elevations in diastolic blood pressure, heart rate, and core body temperature, as well as significant delays in reaction time and diminished working memory performance.

The application of continuous performance testing (CPT) and multi-level n-back tasks enabled a comprehensive evaluation of sustained attention and executive cognitive capacity. The findings indicate that co-exposure to environmental stressors impairs cognitive efficiency to a greater extent than single-stressor scenarios, reinforcing the emerging consensus on the interactivity of physical workplace hazards.

Given the methodological rigor of this study—including a randomized crossover design, standardized physiological measurements, and tightly controlled exposure settings—these results highlight the need for integrated occupational exposure limits (OELs) that consider multi-stressor environments rather than addressing each hazard in isolation. Further investigations in real-world occupational contexts, across broader exposure durations and heterogeneous worker populations, are warranted to improve ecological validity and inform evidence-based policy and intervention strategies.

Data availability

The datasets used and/or analyzed during the current study are available upon reasonable request.

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Author contributions

All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by [Siavash Etemadinezhad], [Navid Jamchi] and [Raziyeh Yousefinezhad]. The first draft of the manuscript was written by [Mohamad Amrolahi], [Seyed Ehsan Samaei] and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Declarations

Competing interests
The authors declare no competing interests.

Consent for publication
All authors of this study agree to publish this paper in Environmental Science and Pollution Research.

Ethical approval
All experimental procedures were carried out in accordance with relevant guidelines and regulations. The experimental protocols were approved by the Research Ethics Committee of Mazandaran University of Medical Sciences, Iran (IR.MAZUMS.REC.1401.154). URL: https://ethics.research.ac.ir/ProposalCertificateEn.php?id=261444&Print=true&NoPrintHeader=true&NoPrintFooter=true&NoPrintPageBorder=true&LetterPrint=true.

Consent to participate
Informed consent was obtained from all individual participants included in the study.

Clinical trial number
Not applicable.

Additional information
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